TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**zBMS - A battery management platform with novel sensing and acting capabilities for improved performance and safety**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), **Secondary Battery (√)**, Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), **Aerospace/Marine (√),** Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Application and Utilization Technologies	Batteries	Secondary Batteries
Concept of Proposed Technology	zBMS is a breakthrough battery management ecosystem that exploits physics-based models for improved battery analytics and management. The ecosystem is composed of a superior hardware platform and associated software resources. The hardware platform combines advanced sensing, e.g. based on electrochemical impedance spectroscopy or high-frequency time domain analytics, and innovative acting, which includes improved passive or active balancing as well as battery cell switching concepts. The software resources are composed of the embedded code for real-time operation as well as cloud-based system for long-term data storage and additional analytics, also including the extraction of Battery Passport features. The novel system overcomes known shortcomings of current technologies and provides superior performance and safety for highly demanding systems or for extreme operating conditions. The entire ecosystem is designed for safety-critical applications and comes with a pre-certification for either ISO 26'0262 (automotive applications) or IEC 61'508 (safety-critical industrial applications). The use and adaptation for other applications domains, such as stationary energy storage systems (where UL 9540/9540A apply), aerospace (DO-254/DO-178C) or maritime systems, is straightforward. The ecosystem is available as a package or its individual parts.			
Rationale for Joint R&D Project	Modern battery management systems (BMS) face substantial technical shortcomings, including limited early-fault detection, inaccurate state-of-charge (SoC) or state-of-health (SoH) estimation, difficulty managing heterogeneous cell aging, insufficient thermal management, and challenges integrating emerging chemistries. These bottlenecks are increasingly critical for a variety of applications, which push toward higher voltages, faster charging, larger pack sizes, and stricter safety regulations. The development of physical battery models has a long history at the CSEM Battery Innovation Hub [1] and was initially applied to improve SoC estimation, which laid the foundation for the Bestimator™ algorithm framework [2]. The initial work used electrochemical impedance spectroscopy (EIS) signals in combination with distribution of relaxation time (DRT) analysis to determine a physically based electric circuit model (ECM). This award-winning technology [3], [5], [11] was applied successfully to degradation monitoring [4] and SoH estimation [7], as well as safety related aspects such as the estimation of the core temperature of a battery cell [8] or thermal runaway prediction [10]. Besides the sensing part,			

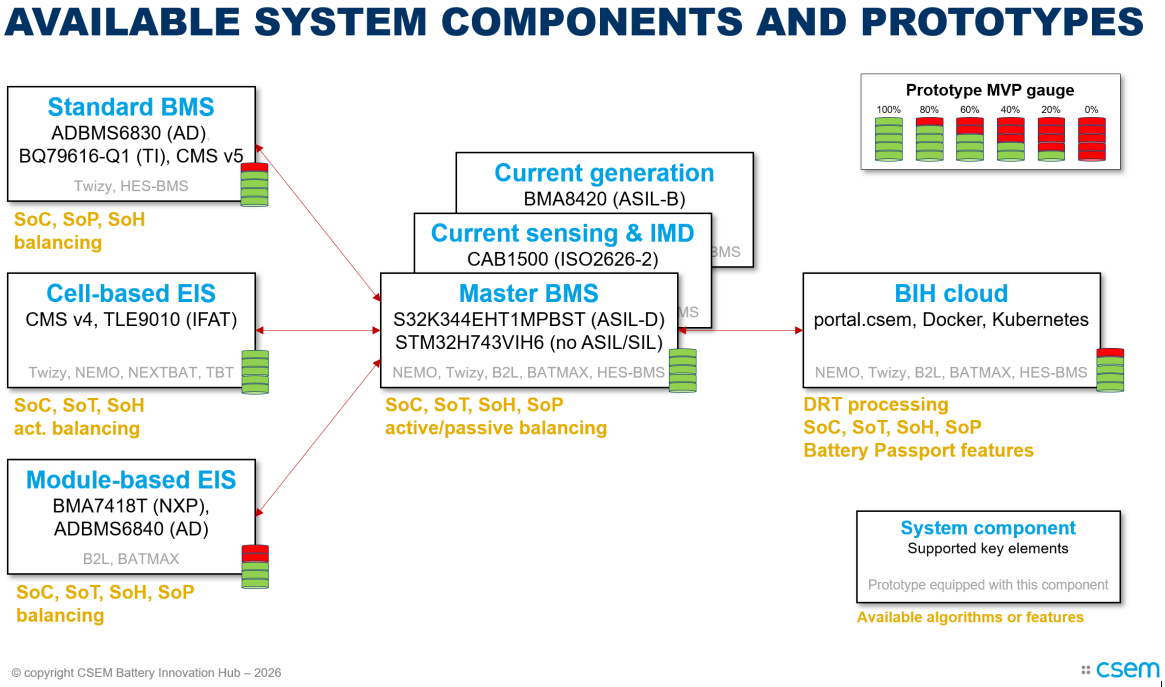
	importance was given to also providing adequate actions, e.g. to ensure balancing all cells in series in a module. The technology was initially developed for NMC chemistries but its applicability to other chemistries, such as LFP, LiM or Na-ion technologies was also shown [6], [9]. Recently, the application of the technology started in Europe, for instance via technology transfer projects like within SMARTBEE [12] or the optimization of the hardware concept that is currently carried out in HES-BMS [13], where the application of pure time-domain analysis (TDA) with fast data acquisition is exploited, especially to also capture long duration electrochemical effects, such as diffusion processes. The patented technology covers hardware concepts as well as algorithmic frameworks [14]-[17] and provides a wide range of applicability for different state estimators and cell technologies. This technology proposal aims at transferring or adapting the technology to Korean SMEs. The offer is aimed at companies that either manufacture or use battery systems and are not satisfied with their current performance. The collaboration consists either of a direct technology transfer of the existing platform and upscaling to meet the required prototype certification or, in the case of new applications, of adapting the necessary hardware and software components to new requirements.
Key Words	1. Physics-based battery cell monitoring 2. In-situ electrochemical impedance monitoring 3. Cell-level active balancing with predictive control 4. Reconfigurable power path and cell-switching topologies 5. Physics-informed digital twins for cell health

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 5-6	End of the Project	Level 7-8

R&D Objectives

The BMS ecosystem has been applied and validated on different prototypes, like for instance in the Renault Twizy demonstrator, which was demonstrated at the CSEM Business Day 2024 [11]. The components of the platform are illustrated in the image below. The hardware components are composed of different front ends (left hand side of the image) that supervise individual or up to 25 cells, the master BMS that comes with auxiliary boards (current sensing, safety functions, current generation) and the optional cloud system. The ecosystem is complemented by associated embedded operational software and the Bestimator™ algorithm framework, which both exploit the features of the hardware platform.



The different parts of this ecosystem can be combined and adapted, thus for operation on systems being composed of single cells, over typical module level systems with several kWh operating typically at 50V up to high-voltage system at e.g. 800V or even 1'500V. The maturity of the different hard- and software components is illustrated in the above picture via the minimal viable product (MVP) gauge and current **R&D development objectives** include the following items:

- Fast voltage and current signal acquisition up to 100kHz for the implementation of the TDA approach, which has been already validated with custom-of-the-shelf components in CMS v5.
- Prototype validation of cost-effective EIS implementation via module-based approach, where commercially available ICs from NXP and AD are used.
- Implementation of the Battery Passport features required for the European Battery Passport, which becomes mandatory as of 2027.

Cooperations in the frame of TIPA R&D projects are intended to exploit or extend the currently available hard- and software

	ecosystem, thus by adapting the available resources to the specific needs of the intended applications and associated specifications of the Korean SME partner. The ambition is to prepare the commercialization stage of new or updated products via adequate reliability testing and prototype certifications. The CSEM Battey Innovation Hub is equipped to execute reliability testing and to also prepare the certification, which will be done in close cooperation with the Korean SME and in accordance with their needs.
R&D Target Specifications	The R&D target specifications depend on the requirements for the specific application envisioned by the Korean SME and will be established on a case-by-case basis together with the Korean SME. An overview of typical performance specifications is provided hereafter as reference. The specifications for the hardware platform typically include the following items: • **Cell voltage measurement accuracy:** Ensures safe operation, reliable balancing, and accurate SoC estimation. Typical measurable specification include accuracy (±1–5 mV for cell voltages between 2.5–4.3 V) and sampling rate: ≥ 1 kHz per channel. • **Current measurement range & accuracy**: Drives thermal protection, current limits, and SoC/SoE calculations. Typical measurable specifications include measurement range (± 300–1000 A), accuracy (±0.5–1% over full operating temperature range) and sampling rate (1–100 kHz). • **Temperature measurement concepts**: Performance and safety rely heavily on thermal monitoring, where standard NTC/PTC based sensing for a reduced number of cells or EIS-based core temperature estimation for all cells can be selected. Typical measurable specifications include accuracy and precision (± 1-2°C for both over the entire operational range from -40°C to 60°C), number of channels (depending on concept) and update rate (typically 0.1 to 10 Hz). • **Balancing concepts**: determines how quickly cell imbalances can be corrected, where improved passive or numerous active balancing concepts or cell-based switching architectures can be selected. Typical measurable specifications include balancing current (from 50–300 mA for passive balancing concepts to several 100 A for switching architectures), balancing resolution (≤10 mV), balancing operating range (only at SoC edges or over the entire voltage range) and balancing power efficiency (>90%). • **Communication interfaces & latency**: Ensures timely transmission of diagnostic and control signals to supervisory controllers. Typical measurable specifications include supported interfaces (CAN, CAN FD, LIN, or SPI), maximum bus latency (≤5–10 ms for safety critical messages) and EMC robustness (compliance with ISO 7637 2, CISPR 25, etc.). The typical specifications for the software components and especially the performance of the most common state estimators are outlined hereafter as follows: • **Fault detection & diagnostic (FDD) latency**: Timely detection of thermal, electrical, or sensor faults is core to functional safety. Typical measurable specifications include over current detection time (≤10 ms), over temperature detection time (≤100 ms), sensor plausibility fault detection time (≤500 ms). • **Algorithm computational efficiency**: Ensures estimators run in real time on embedded computing unit with limited CPU and memory. Typical measurable specifications include maximum algorithm cycle time (≤1–5 ms per iteration), max RAM usage (≤50–300 kB), max program memory (≤200–1000 kB Flash memory). • **SoC estimation accuracy**: SoC is the primary driver for range, charge limits, and user visible battery information. Typical measurable requirements include SoC accuracy (≤ ±2% error over full operating range), convergence time after disturbance (≤ 60 seconds, e.g., after load step or sensor drift correction).

	• **SoH estimation accuracy**: Prevents unexpected degradation, enables predictive maintenance, and governs derating. Typical measurable specifications include capacity estimation error (≤ ±3% over lifetime and for all operating conditions), internal resistance estimation error (≤ ±3%). • **SoP response time & accuracy**: Mirrors the battery's instantaneous ability to accept or deliver current for safety and performance. Typical measurable specifications include SoP update period (≤100 ms), SoP prediction error (≤ ±5% compared to lab characterized model) Besides the hard- and software specifications, target specifications also include the certification targets applicable for the envisioned application.
R&D Plan	The R&D plan is generally adapted to the client's needs, but for safety-critical application we propose to follow the V-model approach. This includes the establishment of safety-by-design elements such as hazard analysis and risk assessment (HARA) or failure modes and effects analysis (DFMEA). These inputs guide the creation of the requirement specifications for the envisioned product and its applications together with the client. The verification of these requirements is typically done via continuous integration testing on unit, component and system level, where model, soft- and hardware-in-the-loop (MIL/SIL/HIL) test rigs that are available at the CSEM Battery Innovation Hub are applied based on suitable design-of-experiment (DoE) configurations that capture the nature of the envisioned applications. For a safety-critical application that follows the V-model approach, the key development phases and tasks, the approximate timelines as well as major deliverables and milestones are typically as follows: • **Phase 1 — Requirements & architecture** (1-2 months) T1.1: Capture system requirements; decompose to HW/SW; define KPIs & acceptance tests. T1.2: Safety planning: HARA, ASIL targets, safety goals; draft safety plan. T1.3: System architecture: pack topology, sensing strategy, balancing concept, comms, diagnostics. T1.4: Plan test assets: cell types, cycler profiles, chamber time, HIL platform, fixtures. **Deliverables**: requirement specifications, safety plan, system architecture, verification plan. **Milestones**: requirements frozen; test plan approved; suppliers and BOM short-list. • **Phase 2 — Modeling, data & algorithms** (1-5 months) T1.1: Cell DoE & characterization: OCV-SOC (multi-T), pulse tests, HPPC, EIS → parameter sets. T2.2: ECM modelling and parameter identification vs. SoC/T/SoH; develop lookup tables with uncertainty bounds. T2.3: SoC/SoH/SoP algorithm design: MIL prototypes, FDD residuals & plausibility logic. T2.4: Calibration & convergence studies: Monte-Carlo robustness; noise tuning; sensor bias/aging robustness. T2.5: SIL implementation: fixed-point feasibility, overflow analysis; MISRA static analysis. **Deliverables**: Model set v1.0, Algorithm Spec, MATLAB/Simulink or C reference, Calibration draft. **Milestones**: SoC and SoP specifications met for MIL validation, FDD latencies met in MIL/SIL. • **Phase 3 — Hardware design & bring-up** (2-5 months) T3.1: Component selection: AFE, ADC, current sensor, microcontroller (ASIL-capable), power supplies, isolation. T3.2: Schematics & layout: precision analogue, EMC practices, thermal design for balance resistors. T3.3: Prototype build (HW1): bench bring-up, power integrity, sensor linearity, balancing function. T3.4: HW2 spin if needed after HW1 learnings; add EMC hardening; design for manufacturability. **Deliverables**: Schematics, PCB (HW1/HW2), BOM, test reports (accuracy, bandwidth, thermal). **Milestones**: measurement specs achieved on bench across temp; balancing spec met; stable comms.

	• **Phase 4 — Integration & verification** (3-9 months) T4.1: HIL platform: plant models + real I/O; full comms stack; scripted regression. T4.2: On-target integration: drivers, RTOS tasks, safety mechanisms (watchdog, CRC, range checks, comms). T4.3: Closed-loop tests: SoC/SoP under drive cycles; induced faults; latency & WCET profiling under load. T4.4: EMC pre-compliance and environmental: surge, EFT/B, radiated/conducted; vibe/thermal cycling of controller. **Deliverables**: HIL test suite, integration report, EMC pre-scan, design validation test report. **Milestones**: all SW/HW specs met in system tests; EMC pre-scan green; safety mechanisms verified. • **Phase 5 — Validation, compliance & industrialization** (Weeks 34–48) T5.1: Field-relevant validation: pack-level tests with real cells/modules; aging/abuse boundary evaluations. T5.2: Safety case & assessments: DFMEA/FMEDA closure; SW safety evidence; traceability to safety goals. T5.3: Manufacturing test strategy: end-of-line (EOL) flashing, calibration routines, boundary tests. T5.4: Documentation & release: user & service manuals, cybersecurity concept. **Deliverables**: Validation report, safety case, EOL test plans, release note. **Milestones**: validation KPIs green; compliance evidence compiled; EOL readiness.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The role of the research institute is to provide innovation content to the common product development, which typically includes the following responsibilities: • **Technology exploration & concept development**: Conduct scientific research, evaluate novel BMS architectures, and build feasibility demonstrators. • **Advanced modelling & algorithm development**: Characterize cells, develop estimation algorithms, and deliver validated simulation/algorithm prototypes. • **Prototype creation & lab validation**: Build research-grade hardware/software prototypes and execute controlled lab and HIL tests. • **Methodology, data, and knowledge transfer**: Provide models, datasets, reports, calibration tools, and training to the industrial partner. • **IP generation & technical support**: Develop new IP, manage background IP boundaries, and support integration with industrial systems.
Expected R&R of Korean SME Partner	The role of the Korean SME partner to provide the product specific needs and constraints for the envisioned application, which typically includes the following responsibilities: • **Product definition & requirement ownership**: Provide market-driven specifications, cost targets, and boundary conditions for the commercial product. • **Industrialization & productization**: Translate prototypes from research institute into manufacturable hardware/software and ensure design for manufacturing, design for test, and reliability. • **Certification & regulatory compliance**: Lead all formal certification processes (EMC, safety, functional safety, cybersecurity, battery standards). • **Manufacturing & supply chain setup**: Establish production processes, sourcing strategy, EOL testing, and supplier qualification. It is envisioned that based on the achieved project results, the Korean SME partner is in a position to prepare the commercial exploitation and plan the market introduction of the product directly after project completion.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This outlined technology proposal aims to develop a **novel Battery Management System (BMS)** that significantly advances the state of the art in battery monitoring, safety, and performance optimization. Combining the scientific expertise of the CSEM Battery Innovation Hub with the Korean SME's productization capabilities, the project will create innovative hardware and software technologies that enable higher accuracy, improved reliability, and enhanced operational efficiency for next-generation battery applications in mobility, stationary storage, and industrial systems. The RTO will lead the development of **advanced sensing architectures**, high-fidelity **battery models**, and **state-estimation algorithms** capable of achieving <±2% SoC accuracy, robust SoH prediction, and fast power-limit estimation and improved safety via thermal runaway prediction. These innovations will be validated through laboratory experiments, hardware-in-the-loop testing, and prototype demonstrators. The resulting technology package includes validated algorithms, characterization datasets, reference hardware, and integration guidelines—forming a solid scientific foundation for industrial product development. The industrial partner will use these research outcomes to accelerate the creation of a **commercial BMS platform**, integrating innovations into manufacturable electronics and embedded software suitable for industrialization. This collaboration substantially reduces development risk, shortens time-to-market, and strengthens the partner's competitive position by enabling differentiated product performance, improved safety margins, and potentially lower system cost. Beyond direct technological outcomes, the project delivers broader **industrial and economic impact**. It reinforces regional leadership in battery technologies, supports emerging energy and mobility sectors, and contributes to sustainability through improved battery lifetime and resource efficiency. The work also stimulates knowledge transfer, workforce development, and innovation spillovers into related fields such as digital twins, diagnostics, and energy-asset management. Overall, supported projects create a clear pathway from **fundamental research to commercial deployment**, generating tangible technological advancements and delivering high economic and societal value.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization